Computing the Aligned Deformation Gradient $\mathbf{F}_{\text{aligned}}$

Removing Arbitrary Mesh-Triangulation Orientations in First Piola–Kirchhoff and Projected Stress Post-Processing for Square and Triangular Lattices

MTM Lattice-Triangulation Documentation

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Abstract

In a 2D triangulated crystal mesh, each triangular element's raw deformation gradient $\mathbf{F}$ is defined relative to a single fixed canonical parent triangle. Because a unit cell consists of both lower-left/up ($\triangle$) and upper-right/down ($\triangledown$) triangles, and because CGAL Delaunay triangulation assigns counterclockwise vertex indices (v_0, v_1, v_2) starting from an arbitrary corner, the raw deformation gradient $\mathbf{F}$ contains an arbitrary discrete reference-space transformation on the right. While the metric tensor $\mathbf{C} = \mathbf{F}^\mathsf{T}\mathbf{F}$ (after Lagrange reduction), the strain energy $W(\mathbf{C}_{\text{red}})$, and the Cauchy stress $\boldsymbol{\sigma}$ are strictly invariant, the First Piola–Kirchhoff stress $\mathbf{P} = \partial W/\partial \mathbf{F}$ and the Projected Stress $\tau = \mathbf{m}\cdot(\mathbf{Pn})$ carry an explicit reference-space index and therefore flip sign or rotate columns between neighboring triangles. This document explains separately how $\mathbf{F}_{\text{aligned}}$ is formulated for the **Square Lattice** and for the **Triangular (Hexagonal) Lattice** (both in the $\mathbb{Z}^2$ square reference frame and in the native equilateral reference frame), and how the entire post-processing stress pipeline follows directly from $\mathbf{F}_{\text{aligned}}$ without altering any downstream formulas.

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1 Why the Raw Deformation Gradient F Depends on Triangle Labelling

Let a triangular element have current deformed vertex positions $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^2$ ordered counterclockwise ($\det[\mathbf{x}_1 - \mathbf{x}_0,\ \mathbf{x}_2 - \mathbf{x}_0] > 0$). Denoting the two deformed edge vectors emanating from vertex v_0 by

$$\mathbf{e}_1 = \mathbf{x}_1 - \mathbf{x}_0, \qquad \mathbf{e}_2 = \mathbf{x}_2 - \mathbf{x}_0, \qquad \mathbf{E} = \begin{bmatrix}\mathbf{e}_1 & \mathbf{e}_2\end{bmatrix} \in \mathbb{R}^{2\times 2}, \tag{1}$$

and the reference edge matrix of the canonical parent triangle by $\mathbf{X}_{\text{ref}} = [\mathbf{E}_{1,\text{ref}},\ \mathbf{E}_{2,\text{ref}}]$, the raw element deformation gradient computed in `ElementTriangle2D::calculate_deformation_gradient` is

$$\mathbf{F} = \mathbf{E}\,\mathbf{X}_{\text{ref}}^{-1}. \tag{2}$$

In a finite-element mesh with dynamic Delaunay remeshing (`MeshGenerator::createTrianglesFromPoints` in `src/MeshGenerator.cpp`), triangles are destroyed and re-created during plastic avalanches, so individual triangles do not store their own initial node coordinates; every triangle is compared against the **single canonical parent triangle** $\mathbf{X}_{\text{ref}}$ via the fixed shape-derivative matrix `dndx`.

After CGAL constructs the Delaunay triangulation, `MeshGenerator::createTrianglesFromPoints` does **not** keep CGAL's raw vertex order. Instead, it explicitly reorders each triangle's three vertices (v_0, v_1, v_2) in two steps:

1. **`reorderForMaxAngle(v0, v1, v2)`**: Loops over all three corners of the triangle, computes the interior angle at each corner from the dot product of the two outgoing edges, and places the vertex with the **largest interior angle** first as `ordered[0]` (v_0). (If all angles are strictly acute, $< 75°$, and the triangle is an isosceles grain-boundary kite with two equal legs, it places the apex between the two equal legs at `ordered[0]`.)
2. **Counterclockwise (`cross_product > 0`) enforcement**: Keeps `ordered[0]` first and orders the remaining two vertices as (`ordered[0]`, `ordered[1]`, `ordered[2]`) or (`ordered[0]`, `ordered[2]`, `ordered[1]`) so that $\det[\mathbf{e}_1, \mathbf{e}_2] > 0$.

How this vertex-ordering rule interacts with the mesh triangles depends on whether the crystal is a **square lattice** or a **triangular (hexagonal) lattice**.

2 Case 1: The Square Lattice

In a square lattice, the canonical reference triangle is the bottom-left right-isosceles triangle with legs of length $h = a_0$ (or $h = 1$ in normalized units):

$$\mathbf{X}_{\text{ref}}^{\text{sq}} = \begin{pmatrix} h & 0 \\ 0 & h \end{pmatrix}, \qquad (\mathbf{X}_{\text{ref}}^{\text{sq}})^{-1} = \frac{1}{h}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{3}$$

Two discrete geometric ambiguities enter the raw deformation gradient $\mathbf{F} = \frac{1}{h}\mathbf{E}$ before any elastic deformation occurs:

2.1 Square Ambiguity 1: The 4 Discrete 90° Half-Cell Orientations ($\mathbf{R}_0, \mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3$)

A unit square $[0, 1] \times [0, 1]$ has four corners and can be split along either diagonal (\ or /) into four possible right-angled half-cell triangles whose 90° corner sits at $(0, 0)$, $(1, 0)$, $(1, 1)$, or $(0, 1)$. Even

when 'reorderForMaxAngle' places the origin v_0 directly at the 90° corner of each triangle, walking counterclockwise produces four different raw deformation gradients rotated by $0°, 90°, 180°, 270°$:

1. **Bottom-left triangle ($\triangle$)** (90° corner at $v_0 = (0,0) \to (1,0) \to (0,1)$, rotation 0°):

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{R}_0. \tag{4}$$

2. **Bottom-right triangle** (90° corner at $v_0 = (1,0) \to (1,1) \to (0,0)$, rotation 90°):

$$\mathbf{e}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{F} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \mathbf{R}_1. \tag{5}$$

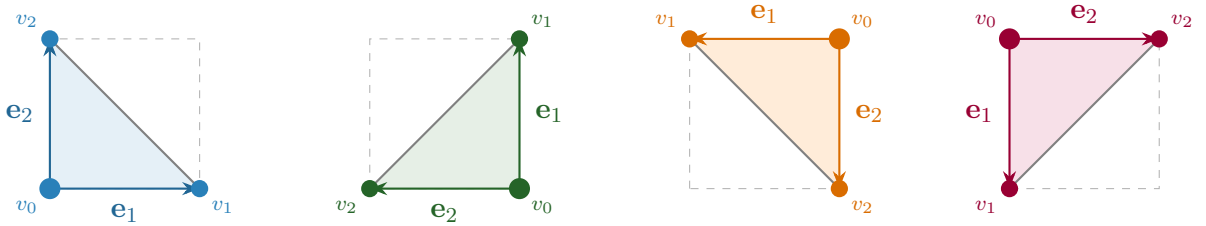
3. **Top-right triangle ($\triangledown$)** (90° corner at $v_0 = (1,1) \to (0,1) \to (1,0)$, rotation 180°):

$$\mathbf{e}_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{F} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \mathbf{R}_2 = -\mathbf{I}. \tag{6}$$

4. **Top-left triangle** (90° corner at $v_0 = (0,1) \to (0,0) \to (1,1)$, rotation 270°):

$$\mathbf{e}_1 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{F} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathbf{R}_3. \tag{7}$$

Right-multiplying $\mathbf{F}$ by $\mathbf{R}_r^\mathsf{T}$ ($r \in \{0,1,2,3\}$) rotates the local reference axes of the triangle by $-0°, -90°, -180°, -270°$, bringing all four half-square triangles back to $\mathbf{F}\mathbf{R}_r^\mathsf{T} = +\mathbf{I}$.



0° ($\mathbf{R}_0$): **Bottom-Left** $v_0 = (0,0)$ $\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

90° ($\mathbf{R}_1$): **Bottom-Right** $v_0 = (1,0)$ $\mathbf{F} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

180° ($\mathbf{R}_2$): **Top-Right** $v_0 = (1,1)$ $\mathbf{F} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

270° ($\mathbf{R}_3$): **Top-Left** $v_0 = (0,1)$ $\mathbf{F} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

Figure 1: **Square Lattice — What the 4 discrete rotations $\mathbf{R}_r$ do:** A unit square $[0,1] \times [0,1]$ (dashed outline) can contain four different right-angled triangle orientations depending on which half of the square the triangle occupies and which diagonal (\ or /) splits the cell. Right-multiplying by $\mathbf{R}_r^\mathsf{T}$ rotates all four orientations back to $+\mathbf{I}$.

2.2 Square Ambiguity 2: The 3 Cyclic Counterclockwise Vertex Shifts $(\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2)$

In an undeformed or weakly deformed square lattice ($90°, 45°, 45°$ triangles), `reorderForMaxAngle` deterministically places v_0 at the 90° corner, so $\mathbf{M}_0 = \mathbf{I}$ is already selected. However, when a triangle is strongly sheared or plastically slipped (for instance, near a dislocation core, across a grain boundary, or when remeshing occurs after finite shear has opened one of the original 45° corners into an obtuse angle $> 90°$), `reorderForMaxAngle` places that obtuse corner at v_0 instead of the original lattice corner. Cycling which of the three corners is chosen as the origin v_0 (while keeping (v_0, v_1, v_2) counterclockwise) transforms the edge matrix $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2]$ by right-multiplication with a unimodular matrix $\mathbf{M}_p \in SL(2, \mathbb{Z})$:

1. **No shift ($\mathbf{M}_0$)**, origin at $v_0 = (0,0) \to v_1 = (1,0) \to v_2 = (0,1)$:

$$[\mathbf{e}_1,\ \mathbf{e}_2] = [\mathbf{e}_1,\ \mathbf{e}_2] \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{\mathbf{M}_0}. \tag{8}$$

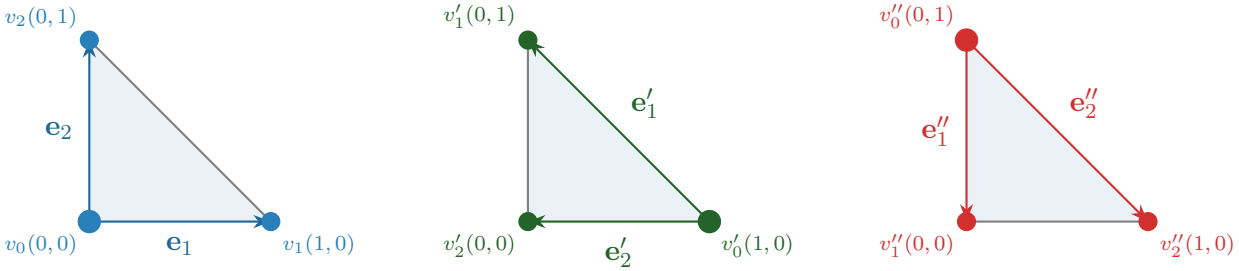
2. **One cyclic shift ($\mathbf{M}_1$)**, origin shifted to $v_0' = (1,0) \to v_1' = (0,1) \to v_2' = (0,0)$: new edges are $\mathbf{e}_1' = \mathbf{x}_2 - \mathbf{x}_1 = -\mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{e}_2' = \mathbf{x}_0 - \mathbf{x}_1 = -\mathbf{e}_1$:

$$[\mathbf{e}_1',\ \mathbf{e}_2'] = [\mathbf{e}_1,\ \mathbf{e}_2] \underbrace{\begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}}_{\mathbf{M}_1}. \tag{9}$$

3. **Two cyclic shifts ($\mathbf{M}_2$)**, origin shifted to $v_0'' = (0,1) \to v_1'' = (0,0) \to v_2'' = (1,0)$: new edges are $\mathbf{e}_1'' = \mathbf{x}_0 - \mathbf{x}_2 = -\mathbf{e}_2$ and $\mathbf{e}_2'' = \mathbf{x}_1 - \mathbf{x}_2 = \mathbf{e}_1 - \mathbf{e}_2$:

$$[\mathbf{e}_1'',\ \mathbf{e}_2''] = [\mathbf{e}_1,\ \mathbf{e}_2] \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}}_{\mathbf{M}_2}. \tag{10}$$

Note that $\det \mathbf{M}_0 = \det \mathbf{M}_1 = \det \mathbf{M}_2 = +1$, and $\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2 \in SL(2,\mathbb{Z})$ form a cyclic group of order 3 ($\mathbf{M}_1^2 = \mathbf{M}_2$, $\mathbf{M}_1^3 = \mathbf{I}$, so $\mathbf{M}_1^{-1} = \mathbf{M}_2$ and $\mathbf{M}_2^{-1} = \mathbf{M}_1$).



No shift ($\mathbf{M}_0$)
Origin at $v_0 = (0,0)$
$\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

1 cyclic shift ($\mathbf{M}_1$)
Origin at $v_0' = (1,0)$
$\mathbf{F} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$

2 cyclic shifts ($\mathbf{M}_2$)
Origin at $v_0'' = (0,1)$
$\mathbf{F} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$

Figure 2: **Square Lattice — What the 3 cyclic shifts $\mathbf{M}_p$ do:** All three panels show the **exact same physical triangle** with vertices at $(0,0)$, $(1,0)$, and $(0,1)$. Depending on which of the 3 corners CGAL lists first as v_0, the two edge vectors emanating from that corner change according to $\mathbf{M}_0$, $\mathbf{M}_1$, or $\mathbf{M}_2$. Multiplying on the right by $\mathbf{M}_p$ shifts the local origin v_0 back to the 90° corner $(0,0)$.

3 Case 2: The Triangular (Hexagonal) Lattice

In a triangular (hexagonal) lattice with nearest-neighbor spacing a_0, every unit rhombus consists of an "up" equilateral triangle ($\triangle$) and a "down" equilateral triangle ($\triangledown$). Unlike the square lattice (where each right triangle has a unique 90° corner that `reorderForMaxAngle` can identify at zero strain), **all three interior angles of an equilateral triangle are equal to** 60°. Consequently, floating-point roundoff or tiny elastic perturbations cause `reorderForMaxAngle` to pick **any of the 3 vertices** as v_0 even in a nearly undeformed crystal!

Combined with the two triangle types ($\triangle$ and $\triangledown$), the counterclockwise pair of edge vectors $(\mathbf{e}_1, \mathbf{e}_2)$ emanating from v_0 can take **6 discrete orientations** rotated by multiples of 60°:

$$\mathbf{E}^{(k)} = \mathbf{R}(k \cdot 60°)\, \mathbf{E}^{(0)}, \qquad k \in \{0,1,2,3,4,5\}, \tag{11}$$

where $k \in \{0, 2, 4\}$ $(0°, 120°, 240°)$ are the 3 cyclic vertex orderings of an "up" triangle ($\triangle$) and $k \in \{3, 5, 1\}$ $(180°, 300°, 60°)$ are the 3 cyclic vertex orderings of a "down" triangle ($\triangledown$).

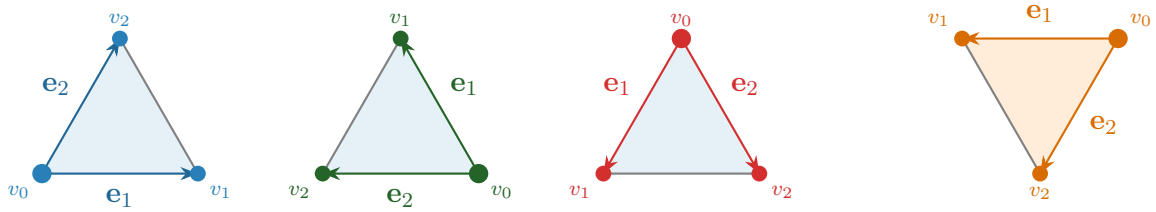
Why Does the Square Lattice Need $4 \times 3 = 12$ Tests, Instead of $6 \times 3 = 18$ in the Triangular Lattice?.

At first sight, since a square has 4 orientations (90° steps) and 3 cyclic vertex orderings ($4 \times 3 = 12$ tests), one might expect a triangular lattice with 6 orientations (60° steps) and 3 cyclic vertex orderings to require $6 \times 3 = 18$ tests. The reason it requires only **6** (and why the 18 products repeat those 6 matrices three times each) is geometric:

- **In the Square Lattice ($4 \times 3 = 12$):** A half-square right triangle has **one 90° corner and two 45° corners** (two legs of length a_0 and one hypotenuse of length $\sqrt{2}a_0$). Because the triangle is *not* equilateral, cycling its 3 vertices $(\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2)$ moves v_0 from the 90° corner to a 45° corner—which is **not a rigid rotation** of the triangle! Thus the 4 rigid 90° rotations of the right triangle $(\mathbf{R}_0, \mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3)$ and the 3 cyclic vertex shifts $(\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2)$ are completely independent, giving $4 \times 3 = 12$ distinct matrices.
- **In the Triangular Lattice ($2 \times 3 = 6$):** A unit 60° rhombus contains only **2 equilateral triangles**: one "up" ($\triangle$, 0°) and one "down" ($\triangledown$, 180°). Because all 3 corners of an equilateral triangle have the exact same 60° angle and side length a_0, **cycling the 3 vertices $(\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2)$ is already a rigid rotation by** $0°, 120°, 240°$:

$$\mathbf{X}_{\text{ref}}^{\text{tri}}\, \mathbf{M}_p\, (\mathbf{X}_{\text{ref}}^{\text{tri}})^{-1} = \mathbf{R}(p \cdot 120°), \qquad p \in \{0, 1, 2\}.$$

  Combining the **2 triangle types** ($\triangle$ at 0° and $\triangledown$ at 180°) with the **3 cyclic vertex orderings** $(0°, 120°, 240°)$ produces the $2 \times 3 = 6$ orientations $(0°, 60°, 120°, 180°, 240°, 300°)$. If you multiply all 6 rotations $\mathbf{R}(k \cdot 60°)$ by the 3 cyclic vertex shifts $\mathbf{R}(p \cdot 120°)$, you get $\mathbf{R}((k + 2p) \cdot 60°)$, which simply permutes the same 6 matrices three times $(18/3 = 6)$.



$\triangle$ $(k = 0,\ 0°)$ $\mathbf{M}_0,\ +\mathbf{I}$ | $\triangle$ $(k = 2,\ 120°)$ $\mathbf{M}_1,\ +\mathbf{I}$ | $\triangle$ $(k = 4,\ 240°)$ $\mathbf{M}_2,\ +\mathbf{I}$ | $\triangledown$ $(k = 3,\ 180°)$ $\mathbf{M}_0,\ -\mathbf{I}$

Figure 3: **Triangular Lattice — Why raw F rotates by multiples of 60°:** Because all interior angles are 60°, CGAL can pick any of the 3 corners of an "up" triangle $\triangle$ $(0°, 120°, 240°)$ or any of the 3 corners of a "down" triangle $\triangledown$ $(180°, 300°, 60°)$ as v_0.

In the MTM codebase (`ReferenceFrame.h` and `ReferenceFrame.cpp`), the triangular lattice can be run in two reference-frame modes, and `computeAlignedDeformationGradient` in `configuration_saver.cpp` handles both:

3.1 Mode 2A: Triangular Lattice in the $\mathbb{Z}^2$ Square Reference Frame (`ReferenceFrame::SquareZ2`, default `-triangular-reference=0`)

In the default $\mathbb{Z}^2$ reference frame, the canonical parent element remains the square right-isosceles triangle $\mathbf{X}_{\text{ref}}^{\text{sq}} = h\,\mathbf{I}$ with $h = a_0/(4/3)^{1/4}$ so that the unit-cell area is normalized to 1. Under the

hexagonal mapping

$$\mathbf{F}_{\text{hex}} = \left(\frac{4}{3}\right)^{1/4} \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}, \qquad \mathbf{C}_{\text{ref}} = \mathbf{F}_{\text{hex}}^{\mathsf{T}} \mathbf{F}_{\text{hex}} = \frac{2}{\sqrt{3}} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \tag{12}$$

the unit square $[0, h] \times [0, h]$ shears into a 60° rhombus consisting of:

- The **lower-left half-square $\hat{\mathcal{T}}_+$ ($\mathbf{R}_0 = +\mathbf{I}$)**, which maps to the "up" equilateral triangle ($\triangle$).
- The **upper-right half-square $\hat{\mathcal{T}}_-$ ($\mathbf{R}_2 = -\mathbf{I}$)**, which maps to the "down" equilateral triangle ($\triangledown$).

Consequently, for any triangle in an undeformed triangular lattice (rotated by θ_0), its raw deformation gradient measured against $\mathbf{X}_{\text{ref}}^{\text{sq}}$ is

$$\mathbf{F} = \pm\, \mathbf{R}(\theta_0)\, \mathbf{F}_{\text{hex}}\, \mathbf{M}_p^{-1}, \qquad p \in \{0, 1, 2\}, \tag{13}$$

where $+1$ ($\mathbf{R}_0$) corresponds to $\triangle$ and -1 ($\mathbf{R}_2$) corresponds to $\triangledown$. Multiplying on the right by $\mathbf{M}_p\, \mathbf{R}_r^{\mathsf{T}}$ with $r \in \{0, 2\}$ (so $\mathbf{R}_r^{\mathsf{T}} = \pm\mathbf{I}$) immediately restores $\mathbf{F}\, \mathbf{M}_p\, \mathbf{R}_r^{\mathsf{T}} = +\mathbf{R}(\theta_0)\, \mathbf{F}_{\text{hex}}$ for every single $\triangle$ and $\triangledown$ triangle in the mesh! (Note that $r \in \{1, 3\}$ corresponds to pairing the two 60° edges with a 90° rotation of the square reference axes, which flips the sign of F_{12} to the obtuse $-1/2$ well.)

3.2 Mode 2B: Triangular Lattice in the Native Equilateral Reference Frame (`ReferenceFrame::Triangular`, `-triangular-reference=1`)

When `-triangular-reference=1` is enabled, the canonical parent triangle in `make_reference_setup` is the equilateral triangle with vertices $p_1 = (0, 0)$, $p_2 = (a_0, 0)$, $p_3 = (\frac{1}{2}a_0, \frac{\sqrt{3}}{2}a_0)$:

$$\mathbf{X}_{\text{ref}}^{\text{tri}} = a_0 \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}, \qquad \mathbf{C}_{\text{ref}} = \mathbf{I}, \qquad \mathbf{F}_0 = \mathbf{I}. \tag{14}$$

Because the reference triangle $\mathbf{X}_{\text{ref}}^{\text{tri}}$ is itself equilateral, the 3 cyclic CCW vertex shifts $\mathbf{M}_p$ and the $\triangle/\triangledown$ sign flip $\pm\mathbf{I}$ conjugate into **exact 60° orthogonal rotations** in the equilateral reference frame:

$$\pm\, \mathbf{X}_{\text{ref}}^{\text{tri}}\, \mathbf{M}_p\, (\mathbf{X}_{\text{ref}}^{\text{tri}})^{-1} = \mathbf{R}(k \cdot 60^\circ) = \begin{pmatrix} \cos(k\pi/3) & -\sin(k\pi/3) \\ \sin(k\pi/3) & \cos(k\pi/3) \end{pmatrix}, \qquad k \in \{0, 1, 2, 3, 4, 5\}. \tag{15}$$

Therefore, in `ReferenceFrame::Triangular`, the raw deformation gradient of an undeformed element is $\mathbf{F} = \mathbf{R}(\theta_0)\, \mathbf{R}(k \cdot 60^\circ)$, and aligning $\mathbf{F}$ simply tests the 6 hexagonal rotations $k \in \{0, 1, 2, 3, 4, 5\}$:

$$\boxed{\mathbf{F}_{\text{aligned}}^{(\text{TriRef})} = \underset{k \in \{0,1,2,3,4,5\}}{\arg\min} \left\| \mathbf{F}\, \mathbf{R}(k \cdot 60^\circ)^{\mathsf{T}} - \mathbf{R}(\theta_0) \right\|_F^2.} \tag{16}$$

4 Unified Formula in $\mathbb{Z}^2$ Reference Mode and Summary Comparison

In the default $\mathbb{Z}^2$ square-reference mode (`ReferenceFrame::SquareZ2`), a crystal in the Zanzotto potential can contain both square and triangular (hexagonal) phases simultaneously (as well as heavily sheared dislocation cores). We therefore use a **single unified formula** in `ReferenceFrame::SquareZ2` that aligns both square and triangular lattices without branching on lattice type.

In the Lagrange fundamental domain $0 \le 2\, C_{12} \le C_{11} \le C_{22}$:

- The **square phase** sits at the left boundary $F_{12} = 0$: $\mathbf{F}_{\text{sq}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.
- The **triangular (hexagonal) phase** sits at the right boundary $F_{12} = +\frac{1}{2}s_0$: $\mathbf{F}_{\text{hex}} = s_0 \begin{pmatrix} 1 & +1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}$.

By placing the target reference matrix $\mathbf{F}_{\text{target}}$ at the **midpoint** ($F_{12} = 1/4$) of the fundamental domain between the square and triangular wells,

$$\mathbf{F}_{\text{target}} = \mathbf{R}(\theta_0) \begin{pmatrix} 1 & 1/4 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{R}(\theta_0) = \begin{pmatrix} \cos\theta_0 & -\sin\theta_0 \\ \sin\theta_0 & \cos\theta_0 \end{pmatrix}, \tag{17}$$

both the square phase ($F_{12} = 0$) and the acute triangular phase ($F_{12} \approx +1/2$) are at distance $\approx 1/4$ from $\mathbf{F}_{\text{target}}$, while the reflected obtuse triangular orientation ($F_{12} \approx -1/2$, $r \in \{1, 3\}$) and unaligned square orientations ($F_{12} = \pm 1$) are at distance $\geq 3/4$.

Combining the 3 cyclic triangle vertex permutations $\mathbf{M}_p$ ($p \in \{0, 1, 2\}$) with the 4 discrete 90° reference rotations $\mathbf{R}_r \in SL(2, \mathbb{Z})$,

$$\mathbf{R}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{R}_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{R}_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{R}_3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{18}$$

yields the unified $\mathbb{Z}^2$-reference formula used for both square and triangular lattices:

$$\boxed{\mathbf{F}_{\text{aligned}}^{(\text{SquareZ2})} = \underset{\substack{p \in \{0,1,2\} \\ r \in \{0,1,2,3\}}}{\arg\min} \left\| \mathbf{F}\, \mathbf{M}_p\, \mathbf{R}_r^{\mathsf{T}} - \mathbf{F}_{\text{target}} \right\|_F^2 .} \tag{19}$$

Lattice & Mode	**Parent Reference $\mathbf{X}_{\text{ref}}$**	**Candidate Transformations**	**Aligned Undeformed $\mathbf{F}_{\text{aligned}}$**
Square Lattice (`SquareZ2`)	Right triangle $h\,\mathbf{I}$	12 candidates in $SL(2, \mathbb{Z})$: $\mathbf{F}\,\mathbf{M}_p\,\mathbf{R}_r^{\mathsf{T}}$ ($p \in \{0..2\}$, $r \in \{0..3\}$)	$\mathbf{R}(\theta_0)\,\mathbf{I}$
Triangular Lattice (`SquareZ2`, default)	Right triangle $h\,\mathbf{I}$, $h = a_0/(4/3)^{1/4}$	Same 12 candidates in $SL(2, \mathbb{Z})$ (acute $\triangle/\triangledown$ select $r \in \{0, 2\}$)	$\mathbf{R}(\theta_0)\,\mathbf{F}_{\text{hex}}$
Triangular Lattice (`Triangular`, `-triangular-reference-a1`)	Equilateral triangle $a_1 \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}$	6 hexagonal rotations in $SO(2)$: $\mathbf{F}\,\mathbf{R}(k \cdot 60°)^{\mathsf{T}}$ ($k \in \{0..5\}$)	$\mathbf{R}(\theta_0)\,\mathbf{I}$

Table 1: Comparison of the $\mathbf{F}_{\text{aligned}}$ calculation across the Square Lattice and the two Triangular Lattice reference-frame modes in `computeAlignedDeformationGradient` (`src/configuration_saver.cpp`).

5 Why Nothing Downstream Needs to Change Once $\mathbf{F}_{\text{aligned}}$ is Found

Drop-In Replacement Property.
Because $\mathbf{F}_{\text{aligned}} = \mathbf{F}\mathbf{N}$ where $\mathbf{N} \in SL(2,\mathbb{Z})$ (in `SquareZ2` mode) or $\mathbf{N} = \mathbf{R}(k \cdot 60°)^{\mathsf{T}}$ (a point-group symmetry of the triangular lattice in `Triangular` mode), $\mathbf{F}_{\text{aligned}}$ belongs to the **exact same lattice symmetry equivalence class** as $\mathbf{F}$. Therefore, replacing $\mathbf{F}$ by $\mathbf{F}_{\text{aligned}}$ at the very start of the stress routine leaves the reduced metric $\mathbf{C}_{\text{red}}$, the strain energy W, and the Cauchy stress $\boldsymbol{\sigma}$ mathematically unchanged, while automatically aligning the First Piola–Kirchhoff stress $\mathbf{P}_{\text{aligned}}$ and Projected Stress τ_{aligned}.

Once $\mathbf{F}_{\text{aligned}}$ is computed via Eq. (16) or Eq. (19), the existing stress pipeline proceeds verbatim:

1. **Aligned Right Cauchy–Green Metric Tensor**:
$$\mathbf{C}_{\text{aligned}} = \mathbf{F}_{\text{aligned}}^{\mathsf{T}}\mathbf{F}_{\text{aligned}} = \mathbf{N}^{\mathsf{T}}\mathbf{F}^{\mathsf{T}}\mathbf{F}\mathbf{N} = \mathbf{N}^{\mathsf{T}}\mathbf{C}\mathbf{N}. \tag{20}$$

2. **Lagrange Reduction to the Fundamental Domain**: In `SquareZ2` mode, applying `lagrange::reduce`($\mathbf{C}_{\text{aligned}}$) finds the unimodular reduction matrix $\mathbf{M} \in GL(2,\mathbb{Z})$ that maps $\mathbf{C}_{\text{aligned}}$ into the unique Lagrange-reduced metric $\mathbf{C}_{\text{red}}$:
$$\mathbf{C}_{\text{red}} = \mathbf{M}^{\mathsf{T}}\mathbf{C}_{\text{aligned}}\mathbf{M}. \tag{21}$$
For bulk elastic elements, $\mathbf{C}_{\text{aligned}}$ is already in the fundamental domain, so $\mathbf{M} = \mathbf{I}$. For plastically slipped elements (or elements at dislocation cores), $\mathbf{M}$ automatically strips the plastic lattice shear! (In `Triangular` reference mode without Lagrange reduction, $\mathbf{M} = \mathbf{I}$ and $\mathbf{C}_{\text{red}} = \mathbf{C}_{\text{aligned}}$.)

3. **Constitutive Energy Derivative**:
$$\mathbf{S}_{\text{red}} = \frac{1}{A_{\text{cell}}}\frac{\partial W(\mathbf{C}_{\text{red}})}{\partial \mathbf{C}_{\text{red}}}. \tag{22}$$

4. **Aligned First Piola–Kirchhoff Stress Tensor**:
$$\boxed{\mathbf{P}_{\text{aligned}} = 2\,\mathbf{F}_{\text{aligned}}\,\mathbf{M}\,\mathbf{S}_{\text{red}}\,\mathbf{M}^{\mathsf{T}}.} \tag{23}$$

5. **Cauchy Stress Tensor (Identical to Unaligned $\boldsymbol{\sigma}$)**:
$$\boxed{\boldsymbol{\sigma}_{\text{aligned}} = \frac{1}{\det \mathbf{F}_{\text{aligned}}}\mathbf{P}_{\text{aligned}}\mathbf{F}_{\text{aligned}}^{\mathsf{T}} = \frac{2}{\det \mathbf{F}_{\text{aligned}}}(\mathbf{F}_{\text{aligned}}\mathbf{M})\,\mathbf{S}_{\text{red}}\,(\mathbf{F}_{\text{aligned}}\mathbf{M})^{\mathsf{T}} = \boldsymbol{\sigma}.} \tag{24}$$
Proof of invariance: Since $\mathbf{C}_{\text{aligned}} = \mathbf{N}^{\mathsf{T}}\mathbf{C}\mathbf{N}$ and $\det \mathbf{N} = 1$, both $\mathbf{C}$ and $\mathbf{C}_{\text{aligned}}$ reduce to the same canonical metric $\mathbf{C}_{\text{red}}$ in the fundamental domain, with $\mathbf{F}_{\text{aligned}}\mathbf{M} = \mathbf{F}\mathbf{M}_{\text{orig}}$ (up to lattice point-group symmetries of W). Hence $(\mathbf{F}_{\text{aligned}}\mathbf{M})\mathbf{S}_{\text{red}}(\mathbf{F}_{\text{aligned}}\mathbf{M})^{\mathsf{T}}$ and $\det \mathbf{F}_{\text{aligned}} = \det \mathbf{F} \det \mathbf{N} = \det \mathbf{F}$ are completely invariant.

6. **Lab-Frame First Piola–Kirchhoff and Projected Stress**: When the crystal is rotated by θ_0 (rotate-and-cut CSL or fixed-BC supercell) and sheared in the lab frame with $\mathbf{F}_{\text{ext}}(\alpha) = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$, rotating the reference index of $\mathbf{P}_{\text{aligned}}$ by $\mathbf{R}(\theta_0)^{\mathsf{T}}$ gives the lab-frame First Piola–Kirchhoff tensor and the work-conjugate projected stress:
$$\mathbf{P}_{\text{lab}} = \mathbf{P}_{\text{aligned}}\,\mathbf{R}(\theta_0)^{\mathsf{T}}, \qquad \boxed{\sigma_{\text{proj}} = \mathbf{P}_{\text{lab}} : \frac{\partial \mathbf{F}_{\text{ext}}}{\partial \alpha} = (\mathbf{P}_{\text{lab}})_{12} = P_{11}^{\text{aligned}} \sin\theta_0 + P_{12}^{\text{aligned}} \cos\theta_0.} \tag{25}$$

6 Worked Numerical Examples for Square and Triangular Lattices

6.1 Square Lattice Example 1: Alternating Bulk Elements (`Element 0` vs. `Element 1`)

The table below shows actual element tensors extracted from a 100×100 square-lattice simulation (`checkpoint_00007.chk` at external stretch $\alpha = 0.1419$). `Element 0` is a lower-left triangle ($\triangle$) whose CGAL vertex order shifted its columns by $\mathbf{M}_2$, while `Element 1` is the adjacent upper-right triangle ($\triangledown$) with opposite orientation:

Quantity	**Element 0 (Square Bulk $\triangle$)**	**Element 1 (Square Bulk $\triangledown$)**
Raw $\mathbf{F}$	$\begin{pmatrix} -1.15035 & -0.00011 \\ 0.00010 & 0.86927 \end{pmatrix}$	$\begin{pmatrix} -1.15039 & 1.15024 \\ 0.00007 & -0.86937 \end{pmatrix}$
Aligned $\mathbf{F}_{\text{aligned}}$	$\begin{pmatrix} 1.15024 & -0.00011 \\ -0.00010 & 0.86927 \end{pmatrix}$	$\begin{pmatrix} 1.15024 & -0.00015 \\ -0.00007 & 0.86930 \end{pmatrix}$
Raw $\mathbf{P}$	$\begin{pmatrix} +0.39109 & -0.00013 \\ +0.00037 & -0.50618 \end{pmatrix}$	$\begin{pmatrix} +0.39137 & -0.00004 \\ -0.00009 & +\mathbf{0.50583} \end{pmatrix}$
Aligned $\mathbf{P}_{\text{aligned}}$	$\begin{pmatrix} -0.39109 & -0.00013 \\ -0.00037 & -\mathbf{0.50618} \end{pmatrix}$	$\begin{pmatrix} -0.39137 & +0.00004 \\ +0.00009 & -\mathbf{0.50583} \end{pmatrix}$
Raw Cauchy $\boldsymbol{\sigma}$	$\begin{pmatrix} -0.44985 & -0.00036 \\ -0.00036 & -0.44001 \end{pmatrix}$	$\begin{pmatrix} -0.45017 & +0.00010 \\ +0.00010 & -0.43972 \end{pmatrix}$
Aligned Cauchy $\boldsymbol{\sigma}_{\text{aligned}}$	$\begin{pmatrix} -0.44985 & -0.00036 \\ -0.00036 & -0.44001 \end{pmatrix}$	$\begin{pmatrix} -0.45017 & +0.00010 \\ +0.00010 & -0.43972 \end{pmatrix}$

Notice two key facts in this table:

- In the raw calculation, P_{22} jumps from -0.50618 on `Element 0` to $+0.50583$ on `Element 1`, and P_{11} has the wrong sign ($+0.391$) relative to σ_{11} (-0.450).
- With $\mathbf{F}_{\text{aligned}}$, both elements have $\mathbf{F}_{\text{aligned}} \approx \text{diag}(1.150, 0.869)$, $\mathbf{P}_{\text{aligned}} \approx \text{diag}(-0.391, -0.506)$ with consistent sign across the entire mesh, and $\boldsymbol{\sigma}_{\text{aligned}} \equiv \boldsymbol{\sigma}$ to machine precision.

6.2 Square Lattice Example 2: Plastically Slipped Elements (`Element 16328` and `Element 16126`)

For triangles that have undergone a plastic slip or sit next to a dislocation core, $\mathbf{F}_{\text{aligned}}$ removes the arbitrary CGAL vertex permutation while Lagrange reduction ($\mathbf{M}$ in Eq. (23)) reduces the slipped metric back to the fundamental elastic domain:

Quantity	Element 16328 (Slipped)	Element 16126 (Slipped)
Raw $\mathbf{F}$	$\begin{pmatrix} -1.14893 & 2.29950 \\ -0.00297 & -0.87059 \end{pmatrix}$	$\begin{pmatrix} 1.15165 & 1.14974 \\ 0.87404 & -0.00331 \end{pmatrix}$
Aligned $\mathbf{F}_{\text{aligned}}$	$\begin{pmatrix} 1.15057 & -1.14893 \\ -0.87355 & -0.00297 \end{pmatrix}$	$\begin{pmatrix} 1.14974 & -1.15165 \\ -0.00331 & -0.87404 \end{pmatrix}$
Raw $\mathbf{P}$	$\begin{pmatrix} +0.38651 & -0.02527 \\ +0.02731 & +0.49104 \end{pmatrix}$	$\begin{pmatrix} -0.02266 & -0.37120 \\ -0.50378 & +0.02636 \end{pmatrix}$
Aligned $\mathbf{P}_{\text{aligned}}$	$\begin{pmatrix} -0.36124 & -0.38651 \\ +0.51835 & -0.02731 \end{pmatrix}$	$\begin{pmatrix} -0.37120 & +0.02266 \\ +0.02636 & +0.50378 \end{pmatrix}$
Raw Cauchy $\boldsymbol{\sigma}$	$\begin{pmatrix} -0.41502 & +0.03097 \\ +0.03097 & -0.45353 \end{pmatrix}$	$\begin{pmatrix} -0.42703 & -0.02985 \\ -0.02985 & -0.44042 \end{pmatrix}$
Aligned Cauchy $\boldsymbol{\sigma}_{\text{aligned}}$	$\begin{pmatrix} -0.41502 & +0.03097 \\ +0.03097 & -0.45353 \end{pmatrix}$	$\begin{pmatrix} -0.42703 & -0.02985 \\ -0.02985 & -0.44042 \end{pmatrix}$

6.3 Triangular Lattice Example: All 6 Mesh Orientations ($\triangle/\triangledown \times 3$ Cyclic Shifts)

Consider a triangular lattice under simple shear $\mathbf{F}_{\text{ext}}(\alpha) = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$ with $\alpha = 0.05$. Depending on whether CGAL picks corner $0, 1$, or 2 as v_0 and whether the element is an "up" ($\triangle$) or "down" ($\triangledown$) triangle, the raw deformation gradient $\mathbf{F}$ takes 6 completely different matrix forms, whereas $\mathbf{F}_{\text{aligned}}$ collapses all 6 to the exact same matrix:

Triangle & CGAL v_0	Raw F in `SquareZ2` Mode	Winning (p^*, r^*)	Aligned $\mathbf{F}_{\text{aligned}}$ (`SquareZ2`)
$\triangle$, corner 0 (0°)	$s_0 \begin{pmatrix} 1 & 0.5 + 0.866\alpha \\ 0 & 0.866 \end{pmatrix}$	$\mathbf{M}_0\, \mathbf{R}_0^{\mathsf{T}}$	$s_0 \begin{pmatrix} 1 & 0.5 + 0.866\alpha \\ 0 & 0.866 \end{pmatrix}$
$\triangle$, corner 1 (120°)	$s_0 \begin{pmatrix} -0.5 + 0.866\alpha & -1 \\ 0.866 & 0 \end{pmatrix}$	$\mathbf{M}_2\, \mathbf{R}_0^{\mathsf{T}}$	$s_0 \begin{pmatrix} 1 & 0.5 + 0.866\alpha \\ 0 & 0.866 \end{pmatrix}$
$\triangle$, corner 2 (240°)	$s_0 \begin{pmatrix} -0.5 - 0.866\alpha & 0.5 - 0.866\alpha \\ -0.866 & -0.866 \end{pmatrix}$	$\mathbf{M}_1\, \mathbf{R}_0^{\mathsf{T}}$	$s_0 \begin{pmatrix} 1 & 0.5 + 0.866\alpha \\ 0 & 0.866 \end{pmatrix}$
$\triangledown$, corner 0 (180°)	$s_0 \begin{pmatrix} -1 & -0.5 - 0.866\alpha \\ 0 & -0.866 \end{pmatrix}$	$\mathbf{M}_0\, \mathbf{R}_2^{\mathsf{T}}$	$s_0 \begin{pmatrix} 1 & 0.5 + 0.866\alpha \\ 0 & 0.866 \end{pmatrix}$
$\triangledown$, corner 1 (300°)	$s_0 \begin{pmatrix} 0.5 - 0.866\alpha & 1 \\ -0.866 & 0 \end{pmatrix}$	$\mathbf{M}_2\, \mathbf{R}_2^{\mathsf{T}}$	$s_0 \begin{pmatrix} 1 & 0.5 + 0.866\alpha \\ 0 & 0.866 \end{pmatrix}$
$\triangledown$, corner 2 (60°)	$s_0 \begin{pmatrix} 0.5 + 0.866\alpha & -0.5 + 0.866\alpha \\ 0.866 & 0.866 \end{pmatrix}$	$\mathbf{M}_1\, \mathbf{R}_2^{\mathsf{T}}$	$s_0 \begin{pmatrix} 1 & 0.5 + 0.866\alpha \\ 0 & 0.866 \end{pmatrix}$

where $s_0 = (4/3)^{1/4} \approx 1.07457$. Similarly, in `ReferenceFrame::Triangular` (`-triangular-reference=1`), the 6 raw element deformation gradients $\mathbf{F} = \mathbf{F}_{\text{ext}}(\alpha)\, \mathbf{R}(k \cdot 60°)$ are right-multiplied by $\mathbf{R}(k^* \cdot 60°)^{\mathsf{T}}$, collapsing all 6 orientations onto the single aligned deformation gradient $\mathbf{F}_{\text{aligned}} = \mathbf{F}_{\text{ext}}(\alpha) = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$ and producing a uniform First Piola–Kirchhoff stress $\mathbf{P}_{\text{aligned}}$ across every $\triangle$ and $\triangledown$ triangle in the mesh.

7 Lattice Orientation φ and Lattice Rotation ω

Independent of $\mathbf{F}_{\text{aligned}}$.
The lattice orientation φ and rotation ω are a separate measure. They do not use $\mathbf{F}_{\text{aligned}}$ or its target matrix: they are computed from the *current* bond vectors of each element and compared with the ideal unit cell, so they do not depend on the reference frame (`SquareZ2` or `Triangular`), on the triangle labelling, on remeshing, or on the starting positions $\mathbf{X}_0$. Code: `measureLatticeOrientation` and `logLatticeRotation` in `src/configuration_saver.cpp`.

7.1 Per element

1. Current edge matrix. With $\mathbf{F} = \mathbf{x}_e \nabla_{\mathbf{X}} \mathbf{N}$ built from the current atom positions, the current edge vectors are

$$\mathbf{E} = \begin{bmatrix} \mathbf{x}_1 - \mathbf{x}_0 & \mathbf{x}_2 - \mathbf{x}_0 \end{bmatrix} = \mathbf{F}\,\mathbf{E}_{\text{ref}}, \qquad \mathbf{E}_{\text{ref}}^{-1} = \begin{bmatrix} \nabla N_1^{\mathsf{T}} \\ \nabla N_2^{\mathsf{T}} \end{bmatrix}. \tag{26}$$

If $\det \mathbf{E} < 0$ (clockwise nodes) the two columns are swapped; elements with $\det \mathbf{F} \le 0$ are discarded.

2. Least-distorted lattice cell. Let $\mathbf{B}$ be the ideal unit cell,

$$\mathbf{B}_{\text{tri}} = \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}, \qquad \mathbf{B}_{\text{sq}} = \mathbf{I}. \tag{27}$$

For every unimodular $\mathbf{M}$ with entries in $\{-1, 0, 1\}$ and $\det \mathbf{M} = 1$ (about 20 matrices, identity first),

$$\mathbf{G} = \mathbf{E}\,\mathbf{M}\,\mathbf{B}^{-1}, \qquad \mathbf{G}_n = \frac{\mathbf{G}}{\sqrt{\det \mathbf{G}}}, \qquad s(\mathbf{M}) = \left\| \mathbf{G}_n^{\mathsf{T}} \mathbf{G}_n - \mathbf{I} \right\|_F^2, \tag{28}$$

and the candidate with the smallest s is kept. Removing the scale makes the choice depend on cell shape only. The distortion $d = \sqrt{s/2}$ is approximately the equivalent shear of the cell (VTK field `ElementLatticeDistortion`).

3. Orientation. The rotation of the polar decomposition $\mathbf{G} = \mathbf{R}\,\mathbf{U}$ has the closed form (2D, $\det \mathbf{G} > 0$)

$$\boxed{\varphi = \operatorname{atan2}(G_{21} - G_{12},\; G_{11} + G_{22}),} \tag{29}$$

defined only modulo the lattice symmetry $\Theta_s = 60^\circ$ (triangular) or 90° (square).

4. Rotation. With θ the starting angle of the element's grain (majority grain label of its three nodes),

$$\boxed{\omega = \operatorname{wrap}_{\Theta_s}(\varphi - \theta)} \tag{30}$$

with $\operatorname{wrap}_{\Theta_s}$ into $(-\Theta_s/2, \Theta_s/2]$ (VTK field `ElementRotationAngleDeg`). A pristine grain gives $\mathbf{G} = \mathbf{R}(\theta)$ exactly, hence $\omega = 0$, in both reference frames.

5. Reported orientation. The VTK field `ElementLatticeOrientationDeg` stores

$$\varphi_{\text{out}} = \theta + \omega, \tag{31}$$

i.e. φ folded around the grain's own starting angle rather than around 0°, so a grain whose starting angle lies near the fold stays continuous (values may lie outside $\pm\Theta_s/2$). Since the orientation is only defined modulo Θ_s, any linear colour scale has a seam; the snapshot viewer therefore shows these fields with a cyclic colour map of period Θ_s, on which $-\Theta_s/2$ and $+\Theta_s/2$ have the same colour.

Elements are excluded from the statistics below when their nodes belong to more than one grain (grain-boundary elements) or when $d > 0.35$ (dislocation cores, phase-transformed regions).

7.2 Per grain and per load step (`lattice_rotation_log.csv`)

With w_e the current element area and $k = 360°/\Theta_s$ (6 triangular, 4 square), the grain mean is the area-weighted circular mean

$$\bar{\omega}_g = \frac{1}{k}\,\text{atan2}\Big(\sum_{e\in g} w_e \sin k\omega_e,\ \sum_{e\in g} w_e \cos k\omega_e\Big), \tag{32}$$

so that values near $\pm\Theta_s/2$ do not cancel. Successive steps are joined, $\bar{\omega}_g^{(n)} \leftarrow \bar{\omega}_g^{(n-1)} + \text{wrap}_{\Theta_s}(\bar{\omega}_g - \bar{\omega}_g^{(n-1)})$, so cumulative rotations beyond $\Theta_s/2$ stay continuous (`Grain`k`RotationDeg`); the current grain orientation is $\varphi_g = \theta_g + \bar{\omega}_g$ (`Grain`k`OrientationDeg`). The rigid rotation of the applied macroscopic deformation is

$$\omega_{\text{applied}} = \text{atan2}\big(F_{21}^{\text{ext}} - F_{12}^{\text{ext}},\ F_{11}^{\text{ext}} + F_{22}^{\text{ext}}\big) \ = \ -\arctan(\gamma/2) \text{ in simple shear}, \tag{33}$$

and `MeanRotationNetDeg` $= \bar{\omega} - \omega_{\text{applied}}$.

Physical meaning. Slip along a lattice plane leaves bond directions unchanged, so dislocation glide alone gives $\omega = 0$; ω measures the turning of the lattice itself (elastic rotation, grain rotation, bending, kinking), as an EBSD map would.

Validation. A crystal rotated by 17° gives $|\omega| < 10^{-9}$ degrees at $\alpha = 0$ (triangular LJ in both reference frames, Zanzotto square); under simple shear the mean follows $-\arctan(\gamma/2)$ to within 10^{-3} degrees, and exactly for a periodic box.